BACHELOR OF SCIENCE WITH HONOURS IN COMPUTING SCIENCE

Final Year Project Report

DRIVE BUDDY - A MACHINE LEARNING-ENHANCED DROWSINESS AND GAZE DISTRACTION MONITORING

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DECLARATION STATEMENT

I certify that the work submitted is my own and that any material derived or quoted from the published or unpublished work of other persons has been duly acknowledged.

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ABSTRACT

Drive Buddy, is a co-pilot safety system employing Eye Aspect Ratio (EAR) Logic and Machine Learning to detect drowsiness through EAR analysis. Using Machine Learning and Image Recognition system, the solution aims detect fatigue markers as well other eye distractions. This study provide details of the conceptual design and basic design for the development of this solution. The study reviews notable research works conducted in the area of EAR analysis, adoption of industry wide thresholds and how to leverage on EAR to measure fatigues. These research studies layout the foundation of this development study and use the research concept, outcomes and challenges to develop a practical solution. Furthermore, the later part of this focuses on the “how” of the solution development came to be, the methods used, and implementation that brings the solution to fruition.

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GLOSSARY

Drive Buddy System - An advanced driver monitoring system designed to detect signs of driver fatigue and enhance road safety through real-time analysis of facial features and driving behavior.

Driver Fatigue Status - The calculation and display of a fatigue score based on multiple indicators, providing the driver with insights into their current fatigue level.

Driving Behavior Scoring - The assignment of points or scores based on positive and negative driving behaviors, offering feedback to encourage safer driving practices.

EAR – Eye Aspect Ratio. A facial feature measurement algorithm that calculates the ratio of distances between specific landmarks on the eyes. Used for eye blink detection and monitoring eye movement.

EAR Threshold - The predefined value used in the EAR algorithm to determine the threshold for blink detection. The threshold may be adjusted based on individual characteristics.

MAR - Mouth Aspect Ratio. A facial feature measurement algorithm focused on the ratio of distances between facial landmarks around the mouth. Applied for detecting yawning as an additional indicator of driver fatigue.

1. CHAPTER 1: INTRODUCTION

In today's rapidly evolving technological landscape, the integration of smart systems into various aspects of daily life has become increasingly prevalent. This project responds to the growing need for innovative solutions in the realm of driver safety. With the rise of smart vehicles and the ever-expanding demands on individuals in transit, ensuring a secure and attentive driving environment is paramount.

While there are numerous research studies conducted on Eye Aspect Ratio (EAR), employing both conventional methods and modern approaches using image recognition software, a notable gap remains in translating these findings into practical solutions. Despite the wealth of knowledge accumulated, there is a lack of tangible products available for consumers. Therefore, the intention of this project is to bridge this gap by leveraging the outputs of research studies to develop a practical and user-friendly product—introducing the proactive and intelligent co-pilot system known as "Drive Buddy".

1.1 PROBLEM STATEMENT

In the early morning of April 25, 1981, Avelino Dizon, a young Pepsi executive was on his way home after a late-night sales plant visit in Tarlac City, Philippines. With only 40 kilometres left in his journey, fatigue took its toll, causing him to drift into sleep behind the wheel. The car veered off course, fell into a ditch and colliding with a tree. Tragically, Avelino lost his life on the scene, leaving an indelible mark on the lives of my wife and her family. Avelino's story joins the grim statistics, underscoring the consequences of drowsy driving incidents.

According to Singapore Medical Journal, driver fatigue is one of the biggest health and safety concerns within the road transport sector (Lim & Chia, 2015). In 2017, US Department of Transportation National Highway Traffic Safety Administration (NHTSA) have estimated 91,000 police-reported crashes involved drowsy drivers. These crashes led to an estimated 50,000 people injured and nearly 800 deaths (NHTSA, 2021).

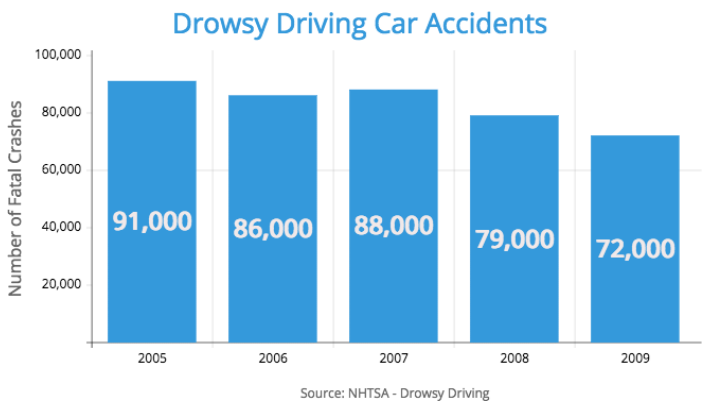


Figure 1.1-1-1 Statistics of Fatal Crashes due to Drowsy Driving (NHTSA, 2021)

The facts of the tragic incident and the statistics from the Singapore Medical Journal and NHTSA, underscore the urgent need for practical solutions to tackle drowsy driving. Taking a proactive stance is crucial to minimize the associated risks and prevent future accidents. It's imperative to develop and implement effective measures, like advanced co-pilot safety systems, to address this significant health and safety issue in the road transport sector. Transforming research findings into practical tools is key to fostering a safer driving environment, ultimately preventing loss of lives and sparing families from the enduring impacts of such incidents.

1.2 SCOPES AND LIMITATIONS

1.2.1 SCOPES

- Design, develop, and implement a co-pilot safety system named "Drive Buddy" using Internet of Things (IoT) solution.
- Incorporate Eye Aspect Ratio (EAR) Logic for real-time monitoring of driver alertness, using Open Computer Vision (OpenCV) and DLIB to measure EAR.
- Utilize machine learning algorithms to enhance drowsiness detection capabilities.
- Implement an intervention system for fatigue management using audio and visual methods, as well as third-party person intervention request.
- Develop a user-friendly mobile application for easy access and interaction and to gamify the driver driving behaviour.

1.2.2 LIMITATIONS

- Effectiveness may be influenced by external factors like diverse driving conditions and individual user behaviours.
- Project scope excludes the physical manufacturing of hardware components, focusing on software and algorithmic development.
- The direct application of certain research outcomes from the EAR studies may have feasibility constraints.
- Despite efforts for continuous improvement, limitations identified during the project may persist in initial iterations.

2. LITERATURE REVIEW

2.1 Overview of Related Work

The referenced research paper, titled "Eye Aspect Ratio for Real-Time Drowsiness Detection to Improve Driver Safety," (Dewi, et al., 2022) addresses the critical issue of drowsiness as a major road safety risk. Investigating the intricate relationship between blink detection and driver safety, the study explores the challenges posed by the rapid blink rate and presents an innovative technique for real-time eye blink identification using a car dashboard camera.

The proposed method involves determining facial landmark positions and extracting the vertical distance between eyelids, utilizing the Eye Aspect Ratio (EAR). This algorithm efficiently estimates facial landmark positions, extracts a scalar quantity using EAR, and identifies eye closeness in each frame. The research establishes the optimal EAR threshold (0.18) through comprehensive experimentation, showcasing improved accuracy in detecting blinks.

The literature review highlights the significance of blink detection technology in various applications, including drowsiness detection, driver safety, computer vision, and face recognition systems. It underlines the importance of maintaining driver concentration during activities such as driving, which influences blink rates. The study also recognizes the impact of factors like eyelid conditions, eye conditions, and environmental stimuli on blink rates.

Several existing models and approaches are discussed, including those incorporating visual context patterns, hierarchical-based interpolation for fatigue indicators, and techniques involving eye closure for blink detection. The work emphasizes the integration of EAR and facial landmarks, drawing on successful methodologies to enhance blink detection accuracy.

2.2 Current State of the Art

The current state of eye blink detection involves a range of methods and techniques that utilize facial landmarks and algorithms for instantaneous analysis.

2.2.1 Eye Aspect Ratio (EAR) Algorithm

The Eye Aspect Ratio algorithm, pioneered by Soukupova and Cech in 2016 (Soukupov´a, 2016), stands as a fundamental component in eye blink detection. By computing the ratio of distances between key facial landmarks, with a primary focus on the eyes, this algorithm proves highly effective in monitoring eye movement and blink patterns. Its simplicity and efficiency have led to widespread adoption in recent research endeavours.

2.2.2 EAR Threshold Variability

Explorations in prior research have delved into the influence of EAR threshold variability on the accuracy of blink detection (Dewi, et al., 2022). Despite the traditional preference for thresholds around 0.2, the current study challenges this norm. Through experimentation with various EAR

thresholds (0.18, 0.2, 0.225, and 0.25), the study emphasizes the need to recalibrate thresholds to enhance accuracy, particularly in scenarios involving distinctive features such as small eyes.

2.2.3 Real-time Applications

In real-time applications, especially in domains like driver safety and drowsiness detection, accurate and prompt blink detection is important. Literature emphasizes that effective blink detection is crucial for averting accidents caused by driver drowsiness (Dewi, et al., 2022). The real-time capabilities demonstrated by the proposed method in the present study align with the increasing demand for efficient systems in practical applications.

2.3 Market Analysis

The global driver drowsiness detection system market is projected to grow substantially, reaching approximately USD 24.4 billion by 2033 from USD 8 billion in 2023, with an expected CAGR of 11.8% between 2024 and 2033 (market.us, 2024). This market segment focuses on preventing accidents caused by driver fatigue, employing sensors and algorithms to monitor behaviors like eye movement and steering patterns. Growth is driven by rising awareness of road safety, stringent regulations, and advancements in AI and machine learning. The market's sustained expansion is expected, especially in regions with strict safety standards, as demand grows for effective measures against the risks of drowsy driving.

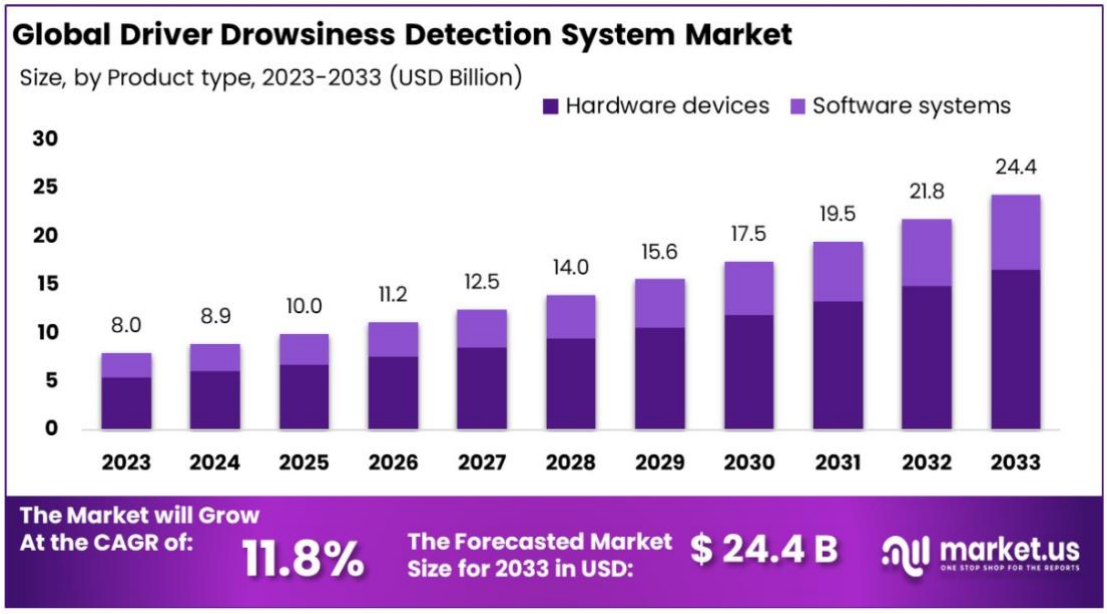


Figure 2.3-1 Global Market Trend of Drowsiness Detection System (market.us, 2024)

2.3.1 Market Players

As per KYTE Research (KYTE Research, 2023), these industry leaders are at the forefront of automotive safety, offering cutting-edge solutions for detecting driver drowsiness. Bosch

utilizes state-of-the-art technology to ensure global safety standards. Smart Eye AB specializes in precise eye-tracking systems, while Seeing Machines Limited provides dependable AI-driven monitoring. Delphi Automotive develops intelligent sensors and algorithms, and Visteon Corporation delivers advanced cockpit solutions with drowsiness detection features. Denso Corporation prioritizes driver well-being with its automotive electronics expertise, and Magna International integrates comprehensive safety solutions into vehicles. Caterpillar Inc. leverages its engineering expertise for advanced driver monitoring, while Takata Corporation focuses on occupant safety with state-of-the-art drowsiness detection systems. Autoliv Inc. offers reliable solutions to mitigate risks associated with driver fatigue, and TRW Automotive enhances road safety through innovative driver assistance systems. Additionally, HELLA contributes to safer roads with advanced driver monitoring solutions. It's important to note that this list is not exhaustive, and there may be other significant players in the automotive safety sector specializing in driver drowsiness detection solutions.

Table 2-3 Key Market Player and Cost

Brand	Product	Estimated Cost (in USD)
Bosch	DriveGuard	$200 - $400
Smart Eye AB	SmartEye Driver Monitor	$500 - $1000
Seeing Machines Ltd	Guardian	$300 - $600
Delphi Automotive	Drowsy Alert System	$150 - $300
Visteon Corporation	DriveSense	$250 - $500
Denso Corporation	Denso Drowsiness Alert	$200 - $400
Magna International	DriverFocus	$300 - $600
Caterpillar Inc.	Cat Driver Safety System	$400 - $800
Takata Corporation	Takata Drowsiness Alert	$250 - $500
Autoliv Inc.	SafeDrive	$350 - $700
TRW Automotive	Alertness Control	$300 - $600
HELLA	HELLA Drowsiness Alert	$200 - $400

3.3.1 Drive Buddy vs Key Market Players Comparison

Drive Buddy's basic device-only product could be priced at approximately $70 commercially. With the addition of a fleet management system, the cost may range from $100 to $120, significantly lower than that of key market players. However, further testing and development are necessary to refine the features for both basic and fleet management tiers to competitively and affordably price the product.

Table 3-3 Key Market Player and Cost

Factors	Drive Buddy	Bosch DriveGuard	SmartEye Driver Monitor	Seeing Machines Guardian
Cost	$70-$120	$200 - $400	$500 - $1000	$300 - $600
Technology	Advanced eye tracking using Machine Learning	Advanced sensor technology, algorithms	Advanced eye-tracking technology	Advanced AI-driven system, eye-tracking
Features	Advanced Eye Tracking, Interventions, alerts using visual and audio	Advanced functionalities, including real-time alerts	Advanced eye-tracking, real-time alerts	Advanced AI-driven monitoring, real-time alerts
Integration	Bluetooth integration with Vehicle	Seamless integration with various vehicle systems	Seamless integration with various vehicle systems	Seamless integration with various vehicle systems
Accessibility	Potential wide accessible	Potentially less accessible	Potentially less accessible	Potentially less accessible
User-Friendliness	Simple interface	User-friendly design	User-friendly design	User-friendly design
Regulatory Compliance	TBD	Meets safety standards and regulations	Meets safety standards and regulations	Meets safety standards and regulations
Reliability	TBD	High reliability and accuracy	High reliability and accuracy	High reliability and accuracy

3. CHAPTER 3: METHODOLOGY

3.1 Research Methods

The study employs research methods that aim to systematically examine and assess the efficacy of the Drive Buddy system in mitigating driver fatigue and improving road safety. These methodologies integrate both quantitative and qualitative strategies, facilitating a thorough exploration of the system's operational effectiveness and user perceptions.

Table 3.1-1 Research Methodologies

#	Phase	Description
1	System Implementation and Configuration	Implement Drive Buddy optimize configuration parameters.
2	Data Collection	Gather data from sensors (eye-tracking, facial recognition, gyroscopes, accelerometers) in various driving scenarios.
3	Machine Learning Model	The models will offer precise facial landmarks, enabling the accurate identification of eye locations for measurement purposes.
4	User Experience Evaluation	Conduct user study to assess driver experiences, feedback on alerts, usability, and satisfaction.
5	Comparative Analysis	Compare Drive Buddy's performance with existing solutions using key metrics (accuracy, false positives, response time).
6	Ethical Considerations	Adhere to ethical guidelines regarding privacy, data security, informed consent, and anonymization.

3.2 Data Collection and Analysis

This section outlines the methodologies employed to collect relevant information and the subsequent analytical techniques applied to derive meaningful insights.

Table 3.2-1 Data Collection Methodologies

Index	Method	Description
1	Sensor-Based Data Collection	Collect data from eye-tracking, facial recognition, gyroscopes, accelerometers in diverse driving scenarios.
2	User Interaction Logging	Log responses to visual and voice alerts, instances of alert triggers, and driver reactions.
3	Questionnaire and Interviews	Administer questionnaires and conduct interviews to gather qualitative feedback on user experience, alert effectiveness, and suggestions for improvement.

4	Facial Recognition Data Anonymization	Anonymize and encrypt facial data for privacy protection.
5	Machine Learning Model Evaluation	Assess model performance using metrics like accuracy, precision, recall. Conduct cross-validation.
6	Driving Behavior Logging	Log speed, acceleration, deviations, and correlate with fatigue events.

3.3 Testing and Validation

To ensure the robustness and reliability of the Drive Buddy system. This section outlines the comprehensive strategies employed to assess the functionality, performance, and security aspects of the system.

Table 3.3-1 Testing and Validation intended for Drive Buddy project

Section	Details
Testing and Validation Overview	Provides an overview of the testing and validation process for the Drive Buddy system.
Testing Objectives	Outlines the specific goals of testing, including:
	Functionality: Verifying core features like drowsiness detection, alerts, and third-party notifications.
	Performance: Evaluating system responsiveness under various conditions and optimizing resource utilization.
	Security: Assessing resilience to cyber threats and identifying potential vulnerabilities.
	Usability: Ensuring a user-friendly interface and positive user experience.
Validation Strategies	Describes the methods used to validate the system's effectiveness, including:
	Scenario-based Testing: Simulating real-world driving scenarios to assess performance.
	User Feedback Collection: Gathering feedback from drivers, administrators, and third-party contacts.
	Continuous Improvement: Implementing an iterative process for ongoing refinement based on feedback and evolving requirements.

4. CHAPTER 4: REQUIREMENTS GATHERING

4.1 Stakeholder Identification and Requirements

Table 4.1-1 Stakeholder Identification and Requirements

Stakeholder	Description	Requirements
Operators (e.g. Drivers)	Individual utilizing the Drive Buddy system.	• Continuous monitoring of EAR in real-time to detect the onset of fatigue. • Continuous tracking of blink rate as an additional indicator of eye fatigue. • Visual alerts triggered if EAR falls below a specified threshold (0.19) with color-coded indicators (Red, Yellow, Green). • Audible warnings activated if EAR maintains a 4-second average below the threshold. • Automatic notification sent for third-party intervention if there are four consecutive 5-second averages below the threshold. • Logging and analysis of EAR and Blink data per minute within the application to identify safe driving patterns.
Operation Manager (e.g. Fleet Managers)	The person overseeing a fleet of Drive Buddy systems	• Capable of monitoring the telemetry data (system health, geographical location, and travel speed) for a fleet of Drive Buddy systems. • Can instantly assess the fatigue levels of Drive Buddy users in real-time. • Receives intervention requests initiated by the Drive Buddy system. • Functions as a secondary third-party intervention provider. • Has the ability to access Drive Buddy system information using an application.
Emergency Contact	Designated emergency contacts	• Can utilize the application to monitor the geographical location and fatigue level data of a designated Drive Buddy system user. • Receives notifications for third-party intervention requests.
Enforcers/Regulators	Government body or industry authorities overseeing road traffic operations and promoting road safety compliance.	• Real-time access to API data of active Drive Buddy Users to determine location, speed and fatigue level data.

4.2 Information and Data Requirements

Table 4.2-1 Information and Data Requirements

Data Type	Description
Telemetry Data	Real-time system health data (cpu utilization, memory utilization, core temperature). Data is provided by Raspberry Pi unit.
Geographical Location and Speed data	Speed data, Latitude, and longitude coordinates of Drive Buddy system. GPS and Speed data are provided by GPS module.
Fatigue Level Data	Real-time EAR and Blink rates measurements to determine fatigue condition. Data to be provided by camera and OpenCV.
User Profile Data	User profiles with personal preferences and settings.
System Alerts and Notifications	Event logs and system-generated notifications.
Historical Data	Records of user journeys, interventions, and system performance.

4.3 Functional Requirements

Table 4.3-1 Functional Requirements

#	Requirement	Description
1	Real-time Telemetry Monitoring	Continuous monitoring of Drive Buddy devices, including sensor status, cpu utilization, memory utilization, core temperature, and connectivity. Timely alerts and notifications for anomalies.
2	Operator Drowsiness Assessment	Real-time assessment of driver fatigue using eye-tracking technology (EAR and blink rates). Triggering visual and audible warnings for fatigue exceeding a predefined threshold. Parameters: EAR Threshold: 0.19 and Below = High Likely, 0.20 to 0.23 Mid, 0.24 and above = Less Likely. Average EAR (AEAR): 4-5 Second Average to react on speed of 60kph (NHTSA, 2022)
3	Location Tracking	Geographical location tracking

4	Emergency Response and Third-Party Intervention	Enabling users to request emergency assistance and notifying designated third-party intervention providers based on specific criteria (3 consecutive High AEAR)
5	User Profiles and Customization	Allowing users to create and manage profiles with personal preferences and emergency contact information. Customizing the user experience based on individual settings and preferences.
6	Data Logging and Analysis	Maintaining historical records of user journeys, interventions, and system performance metrics. Supporting data analysis to identify trends, assess system effectiveness, and optimize performance.
7	Notification and Alert Management	Generating and managing notifications and alerts for system events, interventions, and user-specific information. Ensuring timely, clear, and actionable notifications.
8	Data Privacy and Security Compliance	Adhering to established privacy policies, guidelines, and encryption standards to protect user data and comply with privacy regulations.
9	API Integration and External System Interaction	Providing API endpoints and integration protocols to allow external systems to access real-time Drive Buddy User data for monitoring and analysis.
10	Report Dashboard	Creating a user-friendly report dashboard to visualize and present data analytics results and historical performance metrics.

4.4 Non-Functional Requirements

Table 4.4-1 Non-Functional Requirements

#	Requirement	Actual Requirements
1	Performance Requirements	• The system should respond to user requests within 2 seconds under normal load conditions. • During peak usage hours, the system should maintain a minimum throughput of 1000 requests per minute. • The system's CPU utilization should not exceed 70% during normal operation. • The database should be able to handle a sustained data input rate of at least 1000 records per second.
2	Scalability and Capacity Requirements	• The system should be capable of accommodating a minimum of 1,000 new users per month. • It should support up to 1,000 concurrent users without significant performance degradation. • The data storage capacity should scale automatically to handle at least 5 years of historical data.

3	Availability and Reliability Requirements	• The system should have an uptime of 99.99% or higher, with planned maintenance windows communicated in advance. • In the event of a hardware failure or system crash, the system should recover within 30 minutes. • Data backups should be performed daily, and data restoration should not take longer than 4 hours in case of data loss.
4	Security and Data Privacy Requirements	• User data, including location and fatigue levels, must be encrypted both in transit and at rest. • Access to sensitive data should be restricted to authorized personnel only, following role-based access control. • The system must comply with GDPR regulations regarding the handling of user data.
5	Usability and User Experience Requirements	• The user interface should follow Web Content Accessibility Guidelines (WCAG) to ensure accessibility for users with disabilities. • The system should provide clear and intuitive error messages and instructions.
6	Compatibility and Integration Requirements	• The system should be compatible with the latest versions of popular web browsers (Chrome, Firefox, Safari, Edge). • Compatibility with Android and iOS mobile devices for the Drive Buddy mobile app is required.
7	Performance Monitoring and Analytics Requirements	• The system should have real-time performance monitoring with automated alerts for critical issues. • Performance metrics, such as response times and system load, should be logged and made available for analysis.

4.5 Regulatory and Compliance Requirements

Table 4.5-1 Regulatory and Compliance Requirements

#	Requirement	Description	Related Singapore LTA Rules
1	Traffic Laws Compliance	Adherence to Singapore's traffic laws and regulations	Road Traffic Act (Chapter 276), Traffic Police Department rules
2	Personal Privacy, Data Privacy and Protection	Compliance with Singapore's data protection regulations (e.g., PDPA)	Personal Data Protection Act (PDPA) Inward facing camera regulation (LTA, 2019)
3	User Safety Standards	Ensure features do not distract drivers and meet LTA safety standards	LTA's Road Safety Handbook, Guidelines for Vehicle Navigation Systems In-vehicle information and communication systems (LTA, 2016)
4	Transparency and User Consent	Clear terms, privacy policy, and informed user consent	PDPA, Consumer Protection (Fair Trading) Act

5	Regular Compliance Audits	Ongoing assessments and documentation of compliance efforts	No specific LTA rule, but compliance with LTA guidelines is crucial especially contractual obligations (e.g. LTA Rights to Audit clauses)

5. CHAPTER 5: ANALYSIS

5.1 User Needs Analysis

Table 5.1-1 User Needs Analysis

Persona	Age	Occupation	Daily Routine	Needs and Preferences
Operator 1 Alvin	45	Dance Instructor	Daily commute in heavy traffic and late night engagements	Need a drowsiness detection alerts, visual drowsiness level monitor, audible alerts when drowsiness onset is detected voice reminder if recurring pattern of drowsiness is detected. Prefer to see in the activity status using GUI. Prefer a secure access to his activities. Prefer to see an on-board Visual Alerts, and Audible Alerts.
Operator 2 Joel (diabetic)	51	Solution Architect	Daily commute in city traffic, occasional out of town driving	Need to detect drowsiness onset, receive alerts as intervention. Prefer to see blink trends and drowsiness level in time series to identify time of day drowsiness pattern. Data may contribute to determining correlation of diabetes medication intake and drowsiness onset due to sugar level change. Prefer to be called by third party if drowsiness level is severe.
Emergency Contact Joan	48	Accountant	Working from Home	Need to receive notification if a person is drowsy and in-flight so she can call to intervene. Prefer to use SMS as primary notification and Email as secondary notification.
Operation Manager Mike	55	Transport Service Owner	Scheduling of cars and trucks	Need to identify vehicle/machinery location, operator drowsiness level, and operation duration. Prefer to use a GUI with tables to summarize data.

Traffic Operator Jim	57	Traffic Enforcer	Monitor and enforce traffic rules and regulations	Need to pull data using online with real-time data of in-flight operators in vicinity for proactive intervention. Prefer online and central data access by Traffic enforcer in-house app.

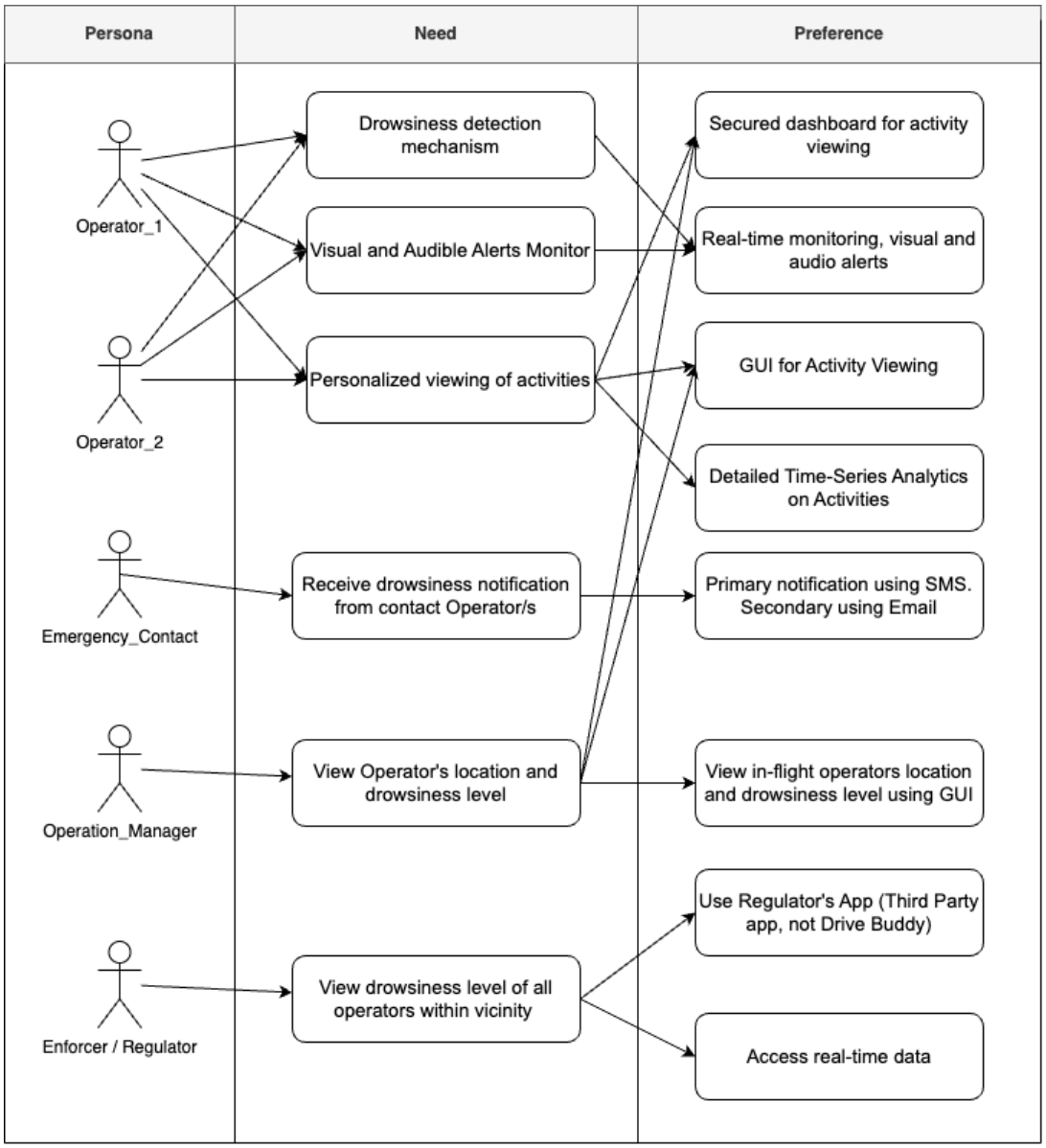


Figure 5.1-1 User Needs Mapping and Rationalization

5.2 Use Case Analysis

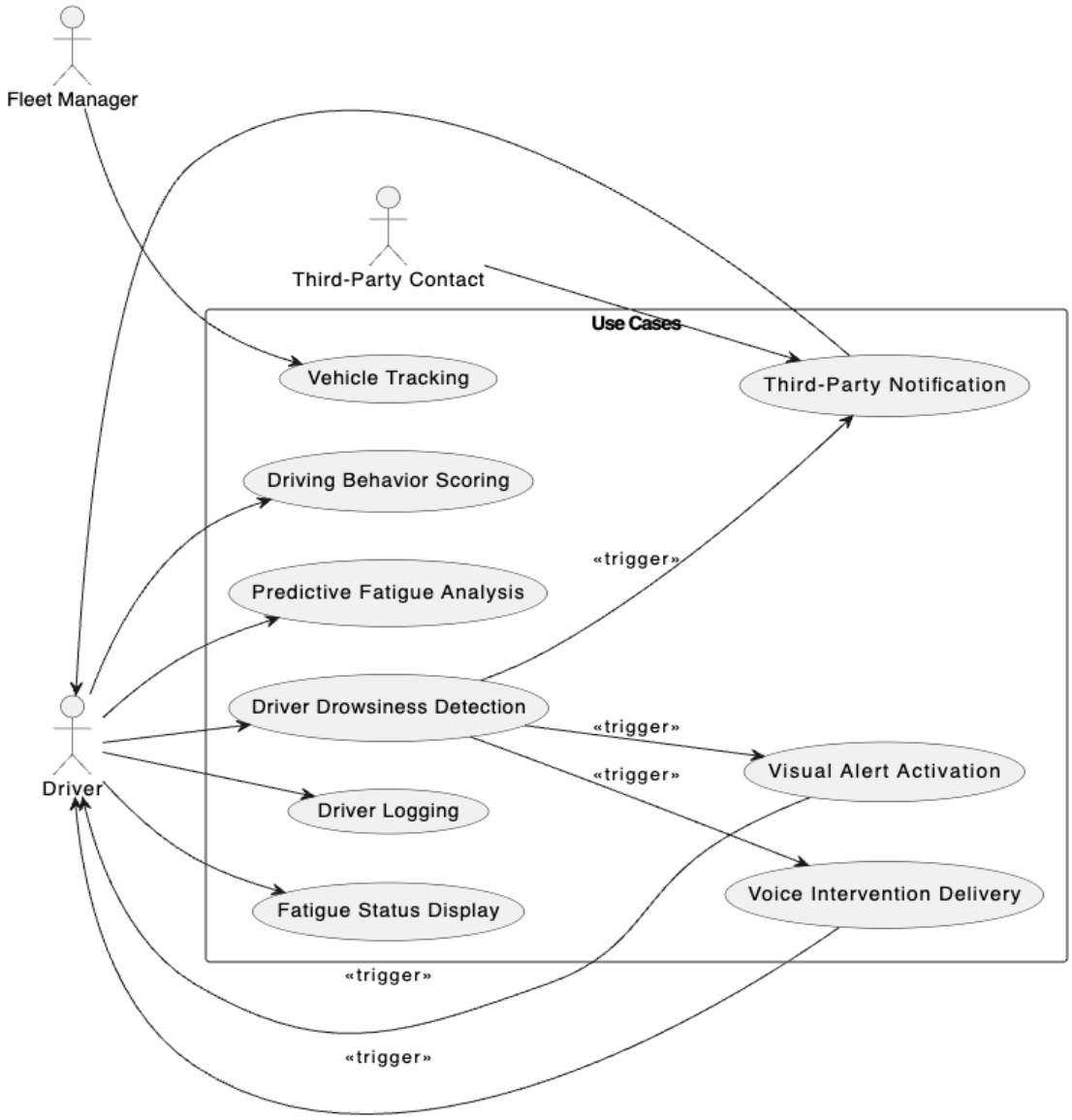


Figure 5.2-1 Use Case Analysis

5.3 Technical Feasibility Fit/Gap Analysis

Table 5.3-1 Technical Feasibility Fit/Gap Analysis

Item	Specs	Fit/Gap
Microcontroller	Raspberry Pi 5 - 8GB RAM	Fit: The 4 ARM Cortex-A76 CPU cores at 2.4GHz and VideoCore VII GPU at 800MHz are suitable for EAR data processing using Python, OpenCV, and DLIB ML Facial Markers for eye marker detection.

		Gap: Standard Debian 12 Bookworm is too resource-intensive for real-time data processing. A lighter Debian 12 version is preferred.
Internet Access	MQTT Transmission to AWS IOT Core	Fit: Singapore's ample bandwidth, similar to many other Asian countries, is generally well-suited for MQTT transmission. The estimated message size for Drive Buddy is relatively small, ranging from 45 to 70 kilobytes. Gap: Challenges arise in areas where access is limited, particularly in some regions.
Camera	Raspberry Pi Camera v3.0	Fit: The Sony IMX708 camera, featuring 11.9 megapixels and a 7.4mm sensor, excels at capturing clear face images. Its narrow 75-degree field of view minimizes distractions, and the F1.8 aperture suits low-light conditions (Raspberry Pi Foundation, 2023). Gap: Unfortunately, this affordable device lacks autofocus and image stabilization which ideal for bumpy roads.
Cloud IoT Service	AWS IoT Core	Fit: Scalable, Secured, Device Management and other IoT management features. Very ideal for large deployments. Gap: Cost and Vendor lock-in is an big concern as most components are using AWS ecosystem.
Machine Learning and Predictive Function	AWS Sagemaker	Fit: Well-suited for backend processing of MQTT messages, with potential for powerful predictions depending on resource allocation, albeit at a potentially high cost for this prototype. Gap: Not suitable for real-time prediction using a 4-second decision split. AWS Sagemaker Neo (Edge ML) is also resource-intensive and may not be suitable for Raspberry Pi.

6. CHAPTER 6: DESIGN

6.1 System Architecture and Components

The Drive Buddy solution comprises two main components: the frontend and backend tiers, leveraging various AWS services to manage, process, and secure telemetry data. The frontend acquires data such as eye tracking, operator identity recognition, speed, geolocation, and other Microcontroller telemetry data.

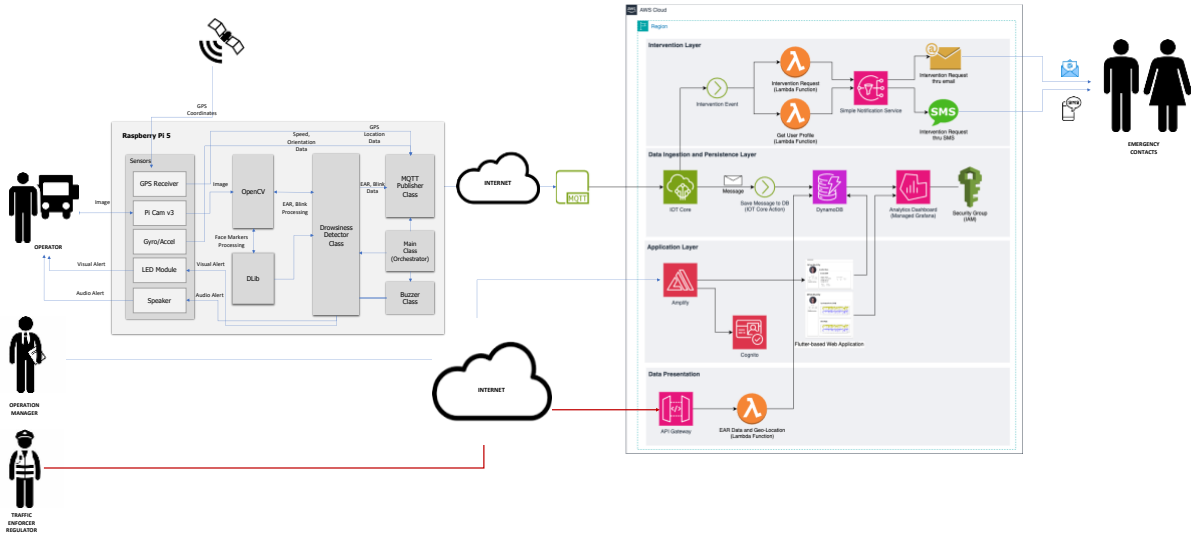


Figure 6.1-1 System Architecture and Components

6.2 User Interface (UI) and User Experience (UX) Design

The solution will employ a multi-channel approach for web and mobile applications to display analytics and manage the Drive Buddy system. Initially, the project aims to design UI/UX for Operators and Operation Managers to view data in near real-time. Prioritizing privacy, each operator will access their profile and data analytics. In enterprise environments like fleet management, Operation Managers will have access to all operators' data analytics.

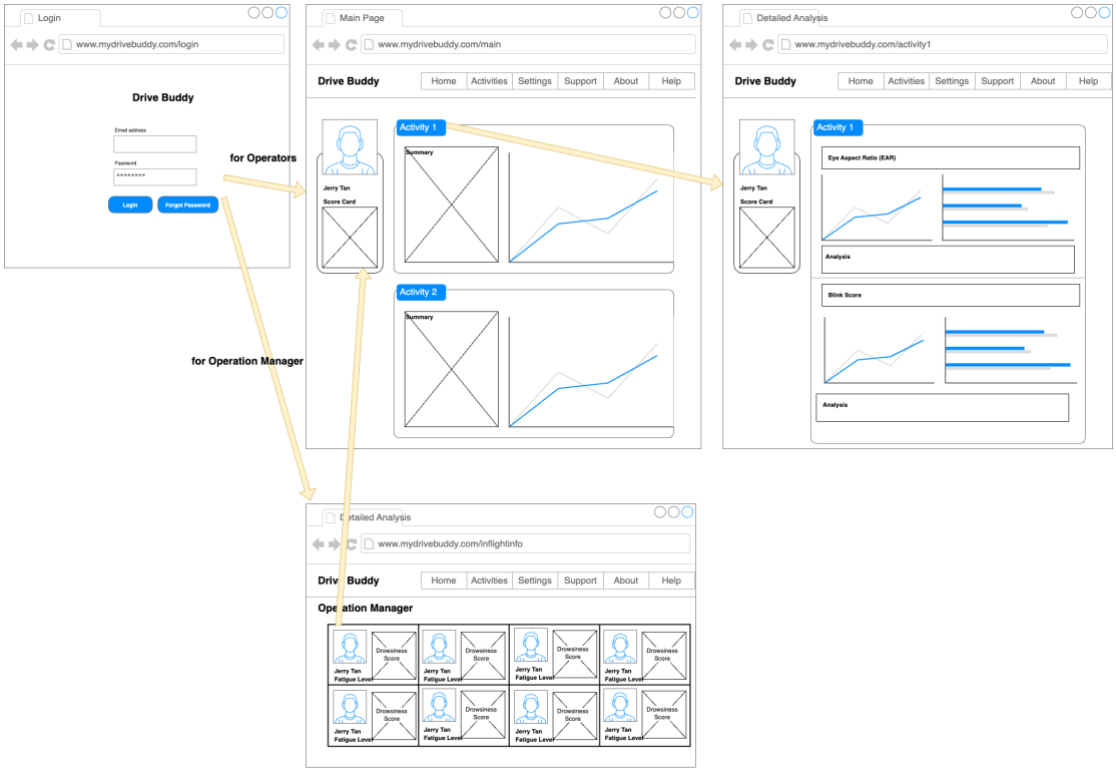


Figure 6.2-1 User Interface and User Experience Design

6.3 Data Modelling and Relationship

Drive Buddy data will be stored in AWS DynamoDB, a NoSQL database. The model illustrates table relationships and the project's approach to normalizing data to its atomic state for efficient storage and retrieval.

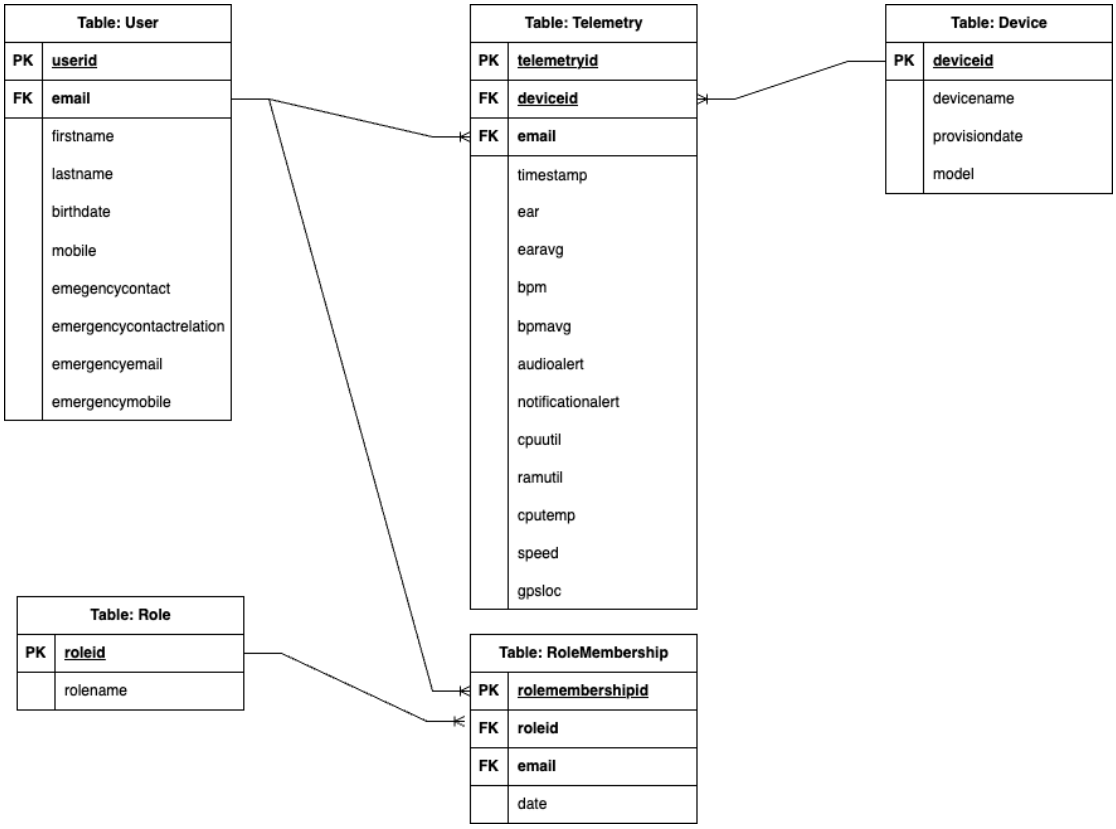


Figure 6.3-1 Entity Relationship Diagram of Drive Buddy

6.4 Security and Privacy Design

Drive Buddy is a critical system designed to save lives by ensuring all components function properly to deliver necessary interventions when needed. Therefore, security and data privacy are top priorities. The security and privacy design outlined below are based on fundamental principles that safeguard the infrastructure, as well as the frontend and backend components.

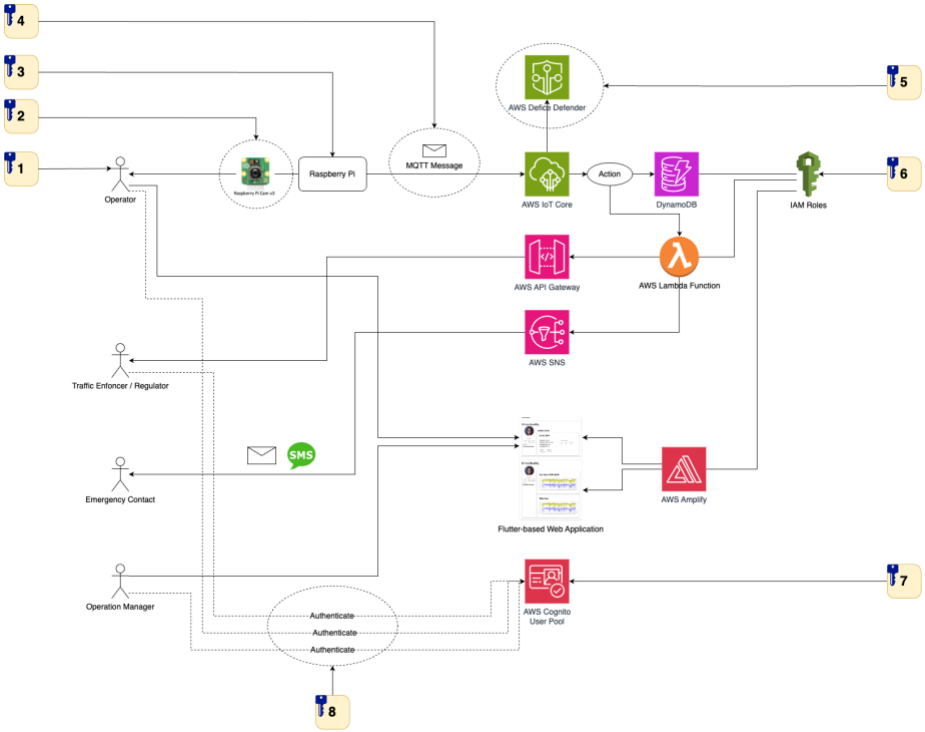


Figure 6.4-1 Security and Privacy Design

Table 6.4-1 Security and Privacy Considerations

#	Threat Vector	Security Control Description
1	Spoofing and Privacy	Biometric access control seamlessly identifies authorized users.
2	Privacy	Uses Pi Cam v3 narrow field of view (75 degrees) to minimize unintended capture of passengers using inward facing camera.
3	Tampering and Unauthorized Access	Hardened Debian 12 Raspberry Pi OS (CIS, 2023)
4	Malicious network impersonators	Utilizes X.509 digital certificates issued individually for each device by AWS IoT Core to safeguard against man-in-the-middle attacks.
5	Unauthorized access, tampering and data breaches	AWS IoT Device Defender is used to detect and mitigate security threats targeting IoT devices (AWS, 2023).
6	Unauthorized access of resources	IAM roles in AWS provide fine-grained access control and permissions management for secure resource access (MICHALOWSKI, 2023).

7	Authentication and Authorization	Cognito user pools manage user authentication and authorization for applications, providing secure access control and identity management (Calandra, 2021).
8	Unauthorized access	Ensure users are validated users and have access to a specific resource (CyberArk, 2023).

7. CHAPTER 7: IMPLEMENTATION

7.1 Development Environment Setup

7.1.1 Sensors and Microcontroller

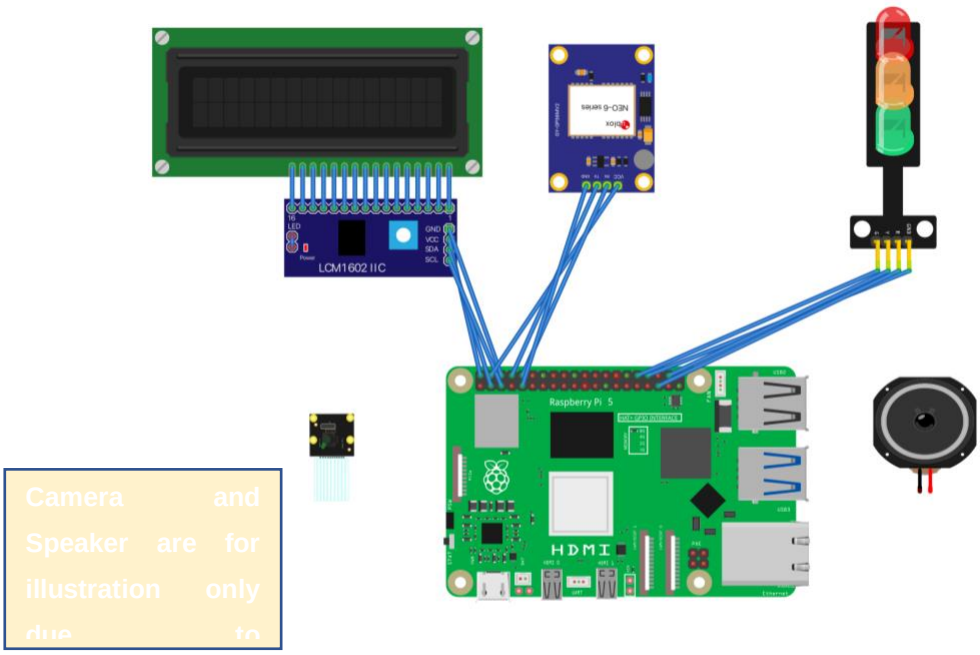


Figure 7.1.1-1 GPIO Pinout Diagram of Sensors and Microcontroller.

Table 7.1.1-1 Security and Privacy Considerations

#	Component	Description
1	Raspberry Pi 5 Board	Debian 12 Bookworm Operating System
2	GPIO Hat Expansion	For easy GPIO connection
3	I2C 16x2 LCD Matrix	Monitor for EAR, AVG EAR, BPM and AVG BPM
4	Seed Speaker Module	Speaker for audio intervention
5	GY-GPS6MU2	GPS receiver to track current location
6	Traffic light Module	For visual monitor average EAR (LOW, MID, HIGH)
7	Raspberry Pi Cam v3	Standard 12MP 75 degree FOV F1.8. Narrow focal optical view with low-light 12mp sensor for eye tracking and blink measurement.

7.1.2 Application Development and Libraries

Table 7.1.2-2 Application Development and Libraries Used

#	Component	Description
1	IDE	Visual Studio Code (for Mac Development, Raspberry Pi, and Flutter)
2	Image Detection	Open Computer Vision (OpenCV) for real-time image detection
3	DLIB	Uses machine learning algorithm to detect 68 facial landmarks. Used for tracking eyes landmark for EAR measurement.
4	Languages	Python – for development in Raspberry Pi Dart – for Flutter development (web application)
5	JSON	For MQTT message format
6	AWS CRT, IOT	For connecting the IOT device to AWS IOT Core.
7	Web Application	Uses Flutter - an open-source UI software development kit created by Google. It is used to develop cross platform applications from a single codebase for any web browser, Fuchsia, Android, iOS, Linux, macOS, and Windows.
8	Hosting Service	AWS Amplify – hosting Flutter application for integrated consumption of AWS services (e.g. AWS API, Lambda, DynamoDB, etc)
9	Code Control	AWS CodeCommit

7.1.3 AWS IOT Core Setup

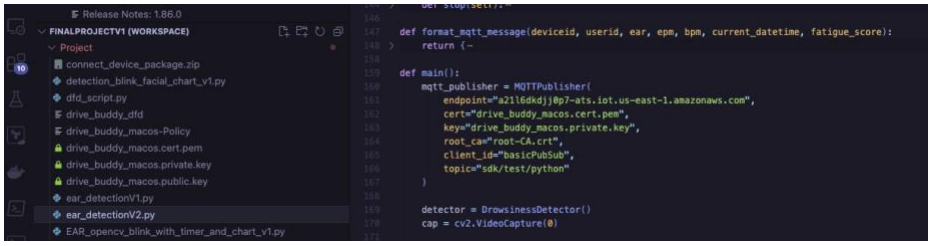


Figure 7.1.3-2 Endpoint connectivity configuration for AWS IOT Core

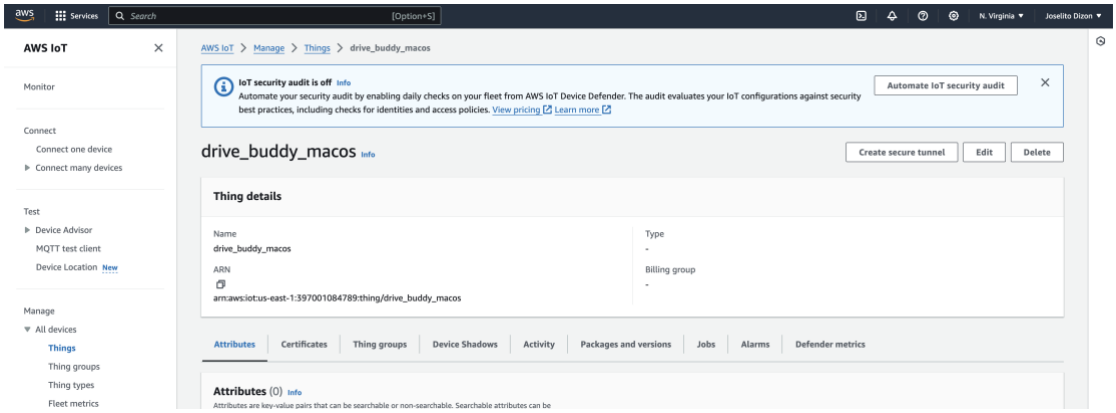


Figure 7.1.3-2 AWS IOT Core Device Configuration

7.1.4 AWS DynamoDB Tables Configuration

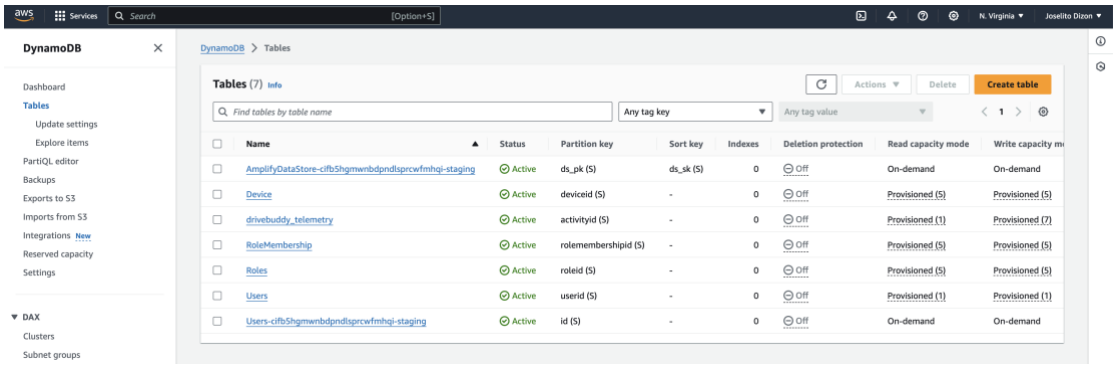


Figure 7.1.4-1 AWS DynamoDB Configuration

7.1.5 AWS Timestream DB Table Configuration

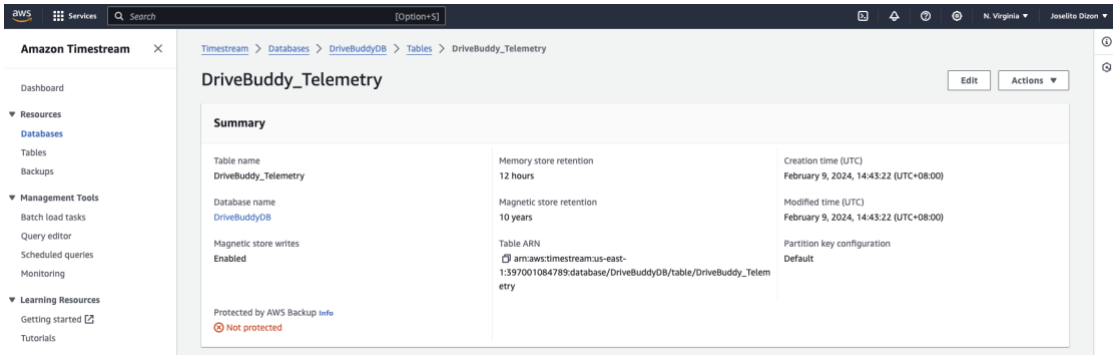


Figure 7.1.5-1 AWS Timestream DB Configuration

7.2 System Development and Coding

7.2.1 Microcontroller and Sensor Setup

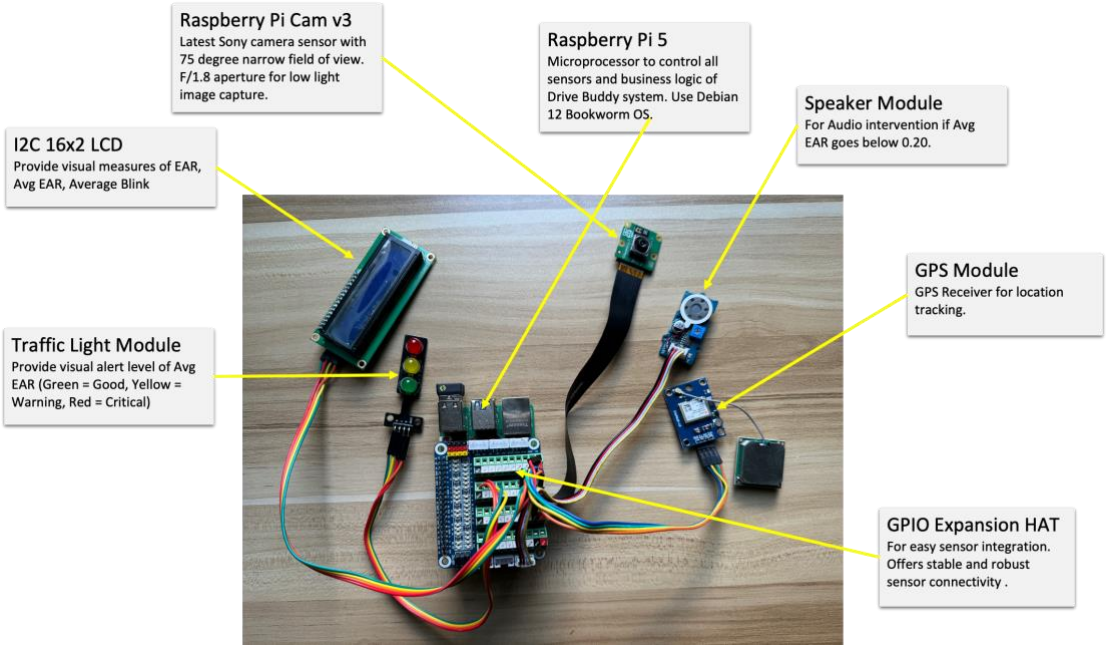
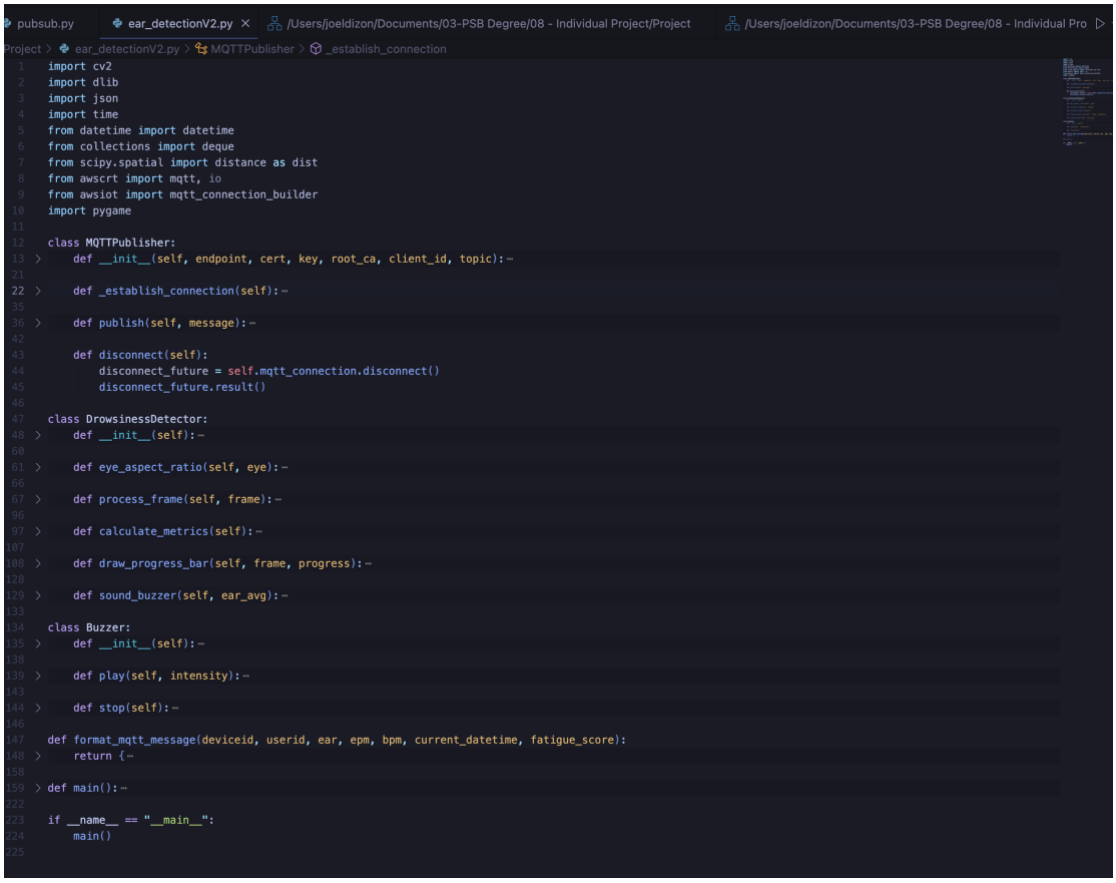


Figure 7.2.1-1 Microcontroller and Sensor Setup

7.2.2 Coding Environment for EAR Detection and Analysis



```
pubsub.py    ear_detectionV2.py ×    /Users/joeldizon/Documents/03-PSB Degree/08 - Individual Project/Project    /Users/joeldizon/Documents/03-PSB Degree/08 - Individual Pro
Project > ear_detectionV2.py > MQTTPublisher > _establish_connection
import cv2
import dlib
import json
import time
from datetime import datetime
from collections import deque
from scipy.spatial import distance as dist
from awscrt import mqtt, io
from awsiot import mqtt_connection_builder
import pygame

class MQTTPublisher:
    def __init__(self, endpoint, cert, key, root_ca, client_id, topic):…

    def _establish_connection(self):…

    def publish(self, message):…

    def disconnect(self):
        disconnect_future = self.mqtt_connection.disconnect()
        disconnect_future.result()

class DrowsinessDetector:
    def __init__(self):…

    def eye_aspect_ratio(self, eye):…

    def process_frame(self, frame):…

    def calculate_metrics(self):…

    def draw_progress_bar(self, frame, progress):…

    def sound_buzzer(self, ear_avg):…

class Buzzer:
    def __init__(self):…

    def play(self, intensity):…

    def stop(self):…

def format_mqtt_message(deviceid, userid, ear, epm, bpm, current_datetime, fatigue_score):
    return {…

def main():…

if __name__ == "__main__":
    main()
```

Figure 7.2.2-1 Classes and Methods Implementation of EAR Detection and Analysis

7.3 Testing and Quality Assurance

7.3.1 Testing EAR and Blink Analysis Code

Drive Buddy has detected an optimal EAR and 4 second average EAR measures. No audio intervention (buzz) has been issued.

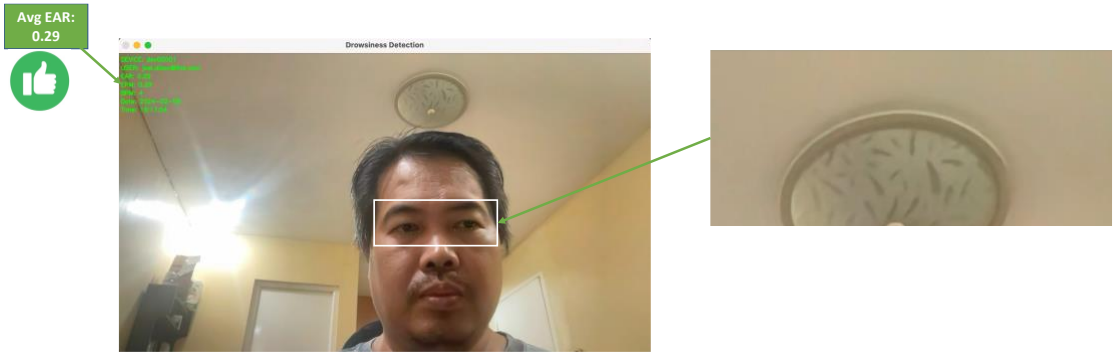


Figure 7.3.1-1 Ideal EAR result and expectation

Drive Buddy has detected a sub-optimal EAR and 4 second average EAR measures. An audio intervention has been alarmed. So, if someone doze off for 4 seconds Drive Buddy will send an alarm. A 4-5 second reaction time according to research is needed to avert a frontal collision for a car running 60kph 4-5 Second Average to react on speed of 60kph (NHTSA, 2022).

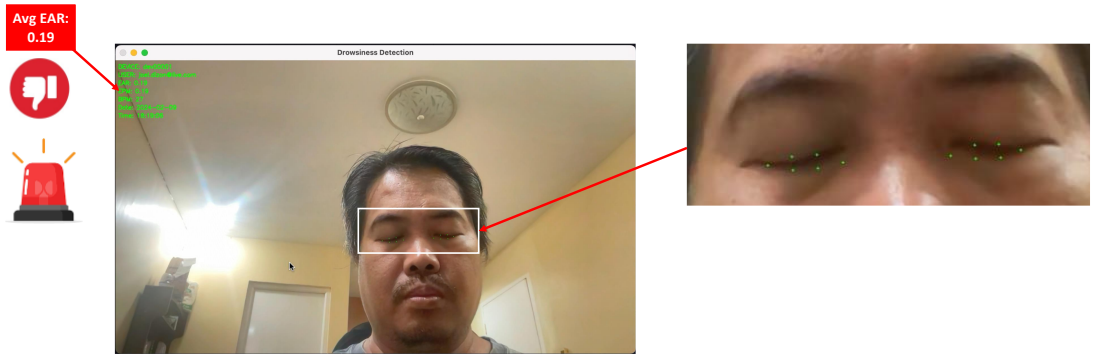


Figure 7.3.1-2 Failed EAR result and expectation

7.3.2 Testing MQTT Message Transmission

Successful transmission of the MQTT message by Drive Buddy IoT device to AWS IOT Core.

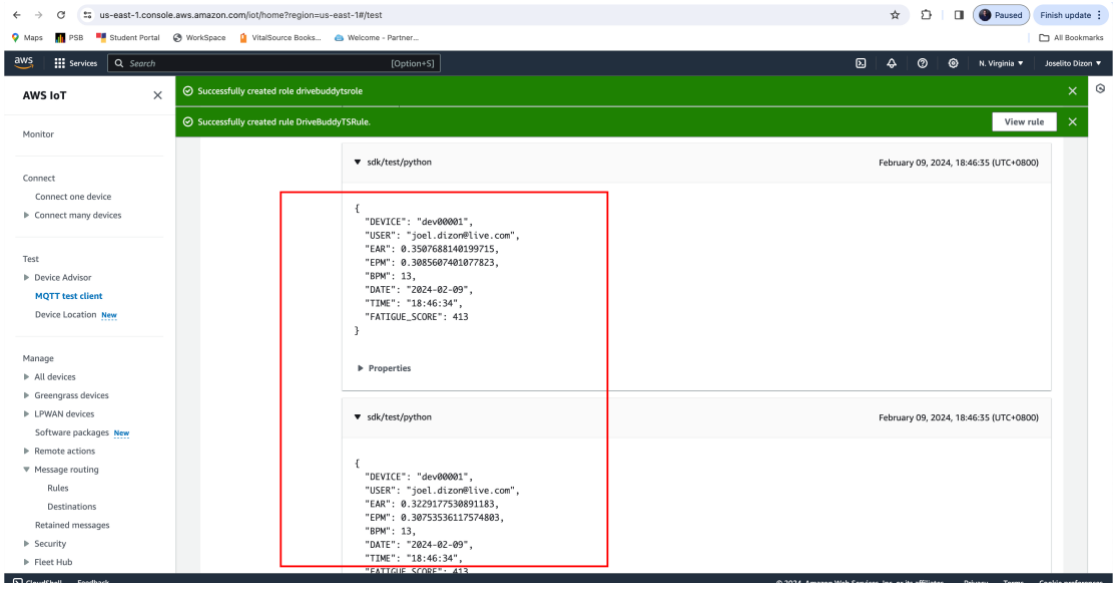


Figure 7.3.2-1 Succesful Real-time MQTT transmission to AWS IOT Core (backend)

7.3.3 Testing MQTT Message Persistence

Successful persistence of MQTT messages to AWS Timestream Database

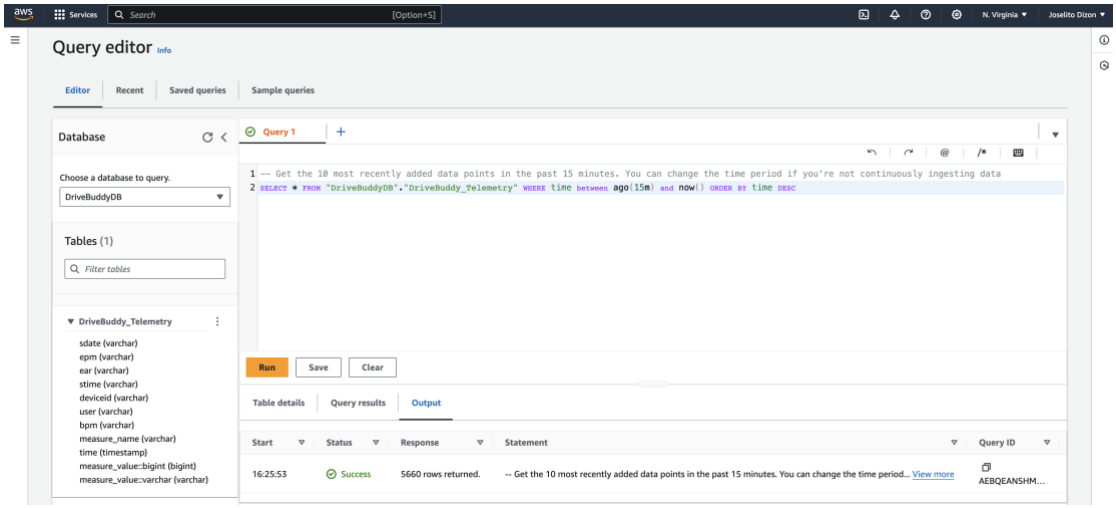
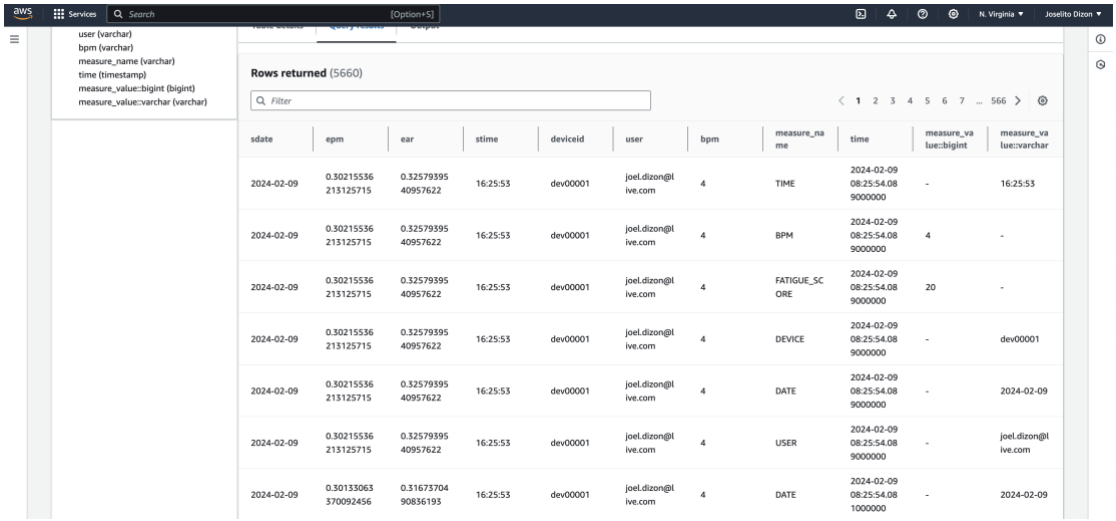


Figure 7.3.3-1 Succesful Persistence of MQTT messages to AWS IOT Core DynamoDB and Timestream DB



sdate	epm	ear	stime	deviceid	user	bpm	measure_name	time	measure_value::bigint	measure_value::varchar
2024-02-09	0.30215536213125715	0.3257939540957622	16:25:53	dev00001	joel.dizon@live.com	4	TIME	2024-02-09 08:25:54.089000000	-	16:25:53
2024-02-09	0.30215536213125715	0.3257939540957622	16:25:53	dev00001	joel.dizon@live.com	4	BPM	2024-02-09 08:25:54.089000000	4	-
2024-02-09	0.30215536213125715	0.3257939540957622	16:25:53	dev00001	joel.dizon@live.com	4	FATIGUE_SCORE	2024-02-09 08:25:54.089000000	20	-
2024-02-09	0.30215536213125715	0.3257939540957622	16:25:53	dev00001	joel.dizon@live.com	4	DEVICE	2024-02-09 08:25:54.089000000	-	dev00001
2024-02-09	0.30215536213125715	0.3257939540957622	16:25:53	dev00001	joel.dizon@live.com	4	DATE	2024-02-09 08:25:54.089000000	-	2024-02-09
2024-02-09	0.30215536213125715	0.3257939540957622	16:25:53	dev00001	joel.dizon@live.com	4	USER	2024-02-09 08:25:54.089000000	-	joel.dizon@live.com
2024-02-09	0.30133063370092456	0.3167370490836193	16:25:53	dev00001	joel.dizon@live.com	4	DATE	2024-02-09 08:25:54.081000000	-	2024-02-09

Figure 7.3.3-2 Succesful Persistence of MQTT messages to DynamoDB and Timestream DB. Payload is distributed to its respective fields.

8. CHAPTER 8: CONCLUSION

The delivery of Drive Buddy final project is a culmination of the knowledge I have acquired in PSB and Coventry University, it signifies a notable milestone, albeit as a prototype. Recognizing its status, there is considerable room for enhancement, with several components yet to be fulfilled.

This project opens up new potential for ongoing improvement and development as it progresses. I acknowledge the need for further discussions to explore avenues for greater improvement such as enhancing user experience, expanding features, ensuring data security, optimizing performance, overcoming challenges due to physical characteristics and nuances (e.g. eye sizes, etc.), and conducting thorough testing. These discussions and refinements are crucial for advancing the project and realizing its full potential.

Moving forward, I am dedicated to refining the prototype, addressing outstanding requirements, and capitalizing on advancements to fully unlock the Drive Buddy solution's potential in fostering road safety and enriching driving experiences.

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APPENDICES